

①_a) $F(x) = 0$ IF $x < 0$, BUT IF $x \geq 0$ THEN

$$F(x) = C \cdot \int_0^x \frac{1}{(u+1)^3} du = C \cdot \left[\left(-\frac{1}{2}\right) \cdot \frac{1}{(u+1)^2} \right]_0^x =$$

$$= C \cdot \frac{1}{2} \cdot \left(1 - \frac{1}{(1+x)^2} \right) \stackrel{\boxed{C=2}}{=} 1 - \frac{1}{(1+x)^2}$$

THUS
 $F(x) \xrightarrow{x \rightarrow \infty} 1$

b) $X = \text{JUMP LENGTH}$ $E(X) = \int_0^\infty 2 \cdot \frac{x}{(1+x)^3} dx =$

$$\int_0^\infty 2 \cdot \frac{x+1}{(1+x)^3} dx - \int_0^\infty 2 \cdot \frac{1}{(1+x)^3} dx =$$

$$2 \cdot \int_0^\infty \frac{1}{(1+x)^2} dx - 2 \cdot \int_0^\infty \frac{1}{(1+x)^3} dx = 2 - 1 = 1$$

$$E(X^2) = \int_0^\infty 2 \cdot \frac{x^2}{(1+x)^3} dx \sim \int_1^\infty \frac{1}{x} dx = +\infty$$

THUS $\text{Var}(X) = +\infty$

c) X_1, X_2, X_3 INDEP. AND IDENTICALLY DISTRIBUTED

$Y := \max\{X_1, X_2, X_3\}$ $G(x) = P(Y \leq x) =$

$$P(X_1 \leq x, X_2 \leq x, X_3 \leq x) = P(X \leq x)^3 = F(x)^3 =$$

$$= \left(1 - \frac{1}{(1+x)^2} \right)^3 \text{ IF } x \geq 0, \text{ OTHERWISE } G(x) = 0$$

P.D.F. OF LONGEST JUMP:

$$g(x) = G'(x) = \frac{d}{dx} (F(x))^3 = 3 \cdot (F(x))^2 \cdot F'(x) = \\ = 3 \cdot \left(1 - \frac{1}{(1+x)^2}\right)^2 \cdot 2 \cdot \frac{1}{(1+x)^3}$$

$$d) P(Y \geq 5) = 1 - G(5) = \\ = 1 - \left(1 - \frac{1}{6^2}\right)^3 = 0.081$$

$$(2) X_m \sim \text{BIN}(m, 0.55)$$

X_m = NUMBER OF STUDENTS WHO PASS

$$\text{WANT: } 0.03 \geq P\left(\frac{X_m}{m} \leq 0.5\right) = P(X_m \leq 0.5 \cdot m) = \\ = P\left(\frac{X_m - (0.55) \cdot m}{\sqrt{(0.55) \cdot (0.45) \cdot m}} \leq \frac{(0.5) \cdot m - (0.55) \cdot m}{\sqrt{(0.55) \cdot (0.45) \cdot m}}\right) \stackrel{\text{DE MOIVRE}}{\approx}$$

$$\approx \Phi\left(\frac{-(0.05)}{\sqrt{(0.55) \cdot (0.45)}} \cdot \sqrt{m}\right) = 1 - \Phi\left((0.1) \cdot \sqrt{m}\right)$$

$$\text{WANT: } \Phi\left((0.1) \cdot \sqrt{m}\right) \geq 0.97 \quad \text{WANT: } \frac{\sqrt{m}}{10} \geq 1.89$$

$$\text{WANT: } m \geq 357.2$$

AT LEAST 358
STUDENTS

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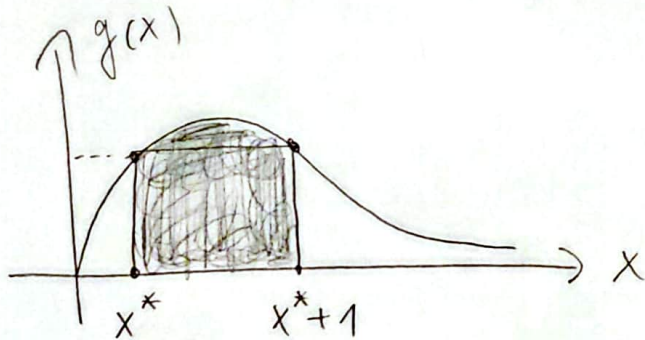
BONUS: UNIT OF MEASUREMENT: MINUTE

Y = TIME OF SECOND CALL

$Y = X_1 + X_2$, WHERE X_1, X_2 ARE
INDEPENDENT $\text{Exp}(\frac{1}{5})$ RANDOM VARIABLES.

P.D.F. OF X_1 IS $\frac{1}{5} \cdot e^{-x/5}$, $\mathbb{I}[0 < x] = f(x)$

P.D.F. OF Y IS $g(x) = (f * f)(x) = \frac{1}{25} \cdot x \cdot e^{-x/5}$



I WANT TO MAXIMIZE
 $\int_{x^*}^{x^*+1} g(x) dx$, SO

CLEARLY, I WANT $g(x^*) = g(x^*+1)$

$$\frac{1}{25} \cdot x \cdot e^{-x/5} = \frac{1}{25} \cdot (x+1) \cdot e^{-(x+1)/5}$$

$$x = (x+1) \cdot e^{-1/5} = x \cdot e^{-1/5} + e^{-1/5}$$

$$x^* = \frac{e^{-1/5}}{1 - e^{-1/5}} = \frac{1}{e^{1/5} - 1} = 0.4516 = x^*$$

THE BEST INTERVAL OF LENGTH 1

$$\text{IS } [x^*, x^*+1]$$

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